

Emilio Hurtado

Only **28.7%** of the seafloor has been mapped to modern standards, yet everything we build at sea starts with understanding what's beneath it. I'm a diver and Marine Sciences student with **field experience in coastal survey and sampling, and data processing in Python and GIS**. I'm set on hydrography because I believe **every project at sea is only as good as the survey behind it**.

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Data acquisition and analysis

I collect my own data **in the field**, then process and analyze it in **Python and GIS**.

Environmental background

I minored in **coastal sustainability** and spent a summer running a **wastewater lab's** routine analyses.

Hackathon winner

Won the 2025 Efiagua Hackathon in Valencia as a **2-person team**, out of **225 participants**.

■ Academic projects

- 2025 ■ **Oceanographic Characterization of the Calpe Coast**
Field campaign · UCV · Calpe, Spain
 - Planned and executed a multi-instrument coastal survey; deployed CTD, multiparameter probe, Niskin bottles and Van Veen grab.
 - Designed side-scan sonar survey lines, acquired the data, and used the imagery to document burial of *Posidonia oceanica* meadows by sand.
 - Responsible for the physical oceanography component: processed CTD, sea-level, chlorophyll and turbidity data in Python (NumPy, pandas, Matplotlib, GSW) to assess Calpe's suitability as a smart coastal city.
- 2025 ■ **Sediment-Sampling Survey Design in the Weddell Sea**
Desk study · UCV
 - Selected a study area and designed a sediment-sampling grid in QGIS based on geological and physical-oceanographic conditions.
- 2024 ■ **Seafloor Geomorphology Mapping in the Alboran Sea**
Desk study · UCV
 - Mapped and interpreted subsea geological structures from public bathymetric datasets in QGIS.
- 2023 ■ **GIS Site Selection for a Wastewater Treatment Plant**
Desk study · UCV
 - Overlaid exclusion layers in ArcGIS (protected land, roads, rivers, livestock routes, rail, towns, steep slopes) to identify a suitable site.

■ Experience

References available upon request

- 2026 ■ **Laboratory Intern**
May–Jul · Ciclagua · EDAR Albufera Sur & EDAR Sueca-Perelló · Valencia, Spain
 - Ran the lab's routine analyses (COD, BOD, suspended solids, nutrients) on influent and effluent samples; results were used for plant process control and regulatory reporting.
- 2024–25 ■ **PADI Divemaster**
Summers · Buceo La Herradura · Granada, Spain
 - Led guided dives and refresher courses, gave dive and boat safety briefings, and ran equipment checks.

Education

BSc in Marine Sciences

Catholic University of Valencia (UCV) · Expected 2027

Minor in Blue Economy and Growth

EU-CONEXUS · Expected 2027

Minor in Coastal Development and Sustainable Maritime Tourism

EU-CONEXUS · 2026

International Baccalaureate

SEK El Castillo, Madrid · 2021
Class representative

International mobilities

SETU, Ireland

Surf & Turf Physics · Oct 2026

Klaipėda University, Lithuania

LNG in shipping · Oct 2026

University of Rostock, Germany

IoT & digital twins, Power BI · May 2026

La Rochelle University, France

Entrepreneurship · Mar 2026

UTCB, Romania

Pollution & remediation · Oct 2024

Languages

Spanish

Native

English

C2 · Cambridge certified

Tools

QGIS

ArcGIS

Python

Git

Linux